Практическое занятие №3

Предел функции

$A = \lim\limits_{\text{x}\to\text{a}} f(x)$, если для $\forall\varepsilon > 0\ \ \exists\ \delta = \delta(\varepsilon) > 0$:

$\forall x$: $\quad 0 < |x - a| < \ \delta => |f(x) - A| < \varepsilon$

Пользуясь определением предела функции, доказать, что $\lim\limits_{\text{x}\to 0} \frac{1}{\text{x}^2+1} = 1$.

$\left|\frac{1}{\text{x}^2+1} - 1\right| = \left|\frac{1-\text{x}^2-1}{\text{x}^2+1}\right| = \frac{\text{x}^2}{\text{x}^2+1} \le \text{x}^2 < \varepsilon \ \ <=> |x| < \sqrt{\varepsilon}, \delta = \sqrt{\varepsilon}.$

$\forall\varepsilon > 0\ \ \exists\ \delta = \delta(\varepsilon) = \sqrt{\varepsilon} > 0 : \forall x: \quad 0 < |x - 0| < \ \delta => \left|\frac{1}{\text{x}^2+1} - 1\right| < \varepsilon =>$

$$\lim_{\text{x}\to 0} \frac{1}{\text{x}^2 + 1} = 1.$$

Вычислить пределы:

1) $\lim\limits_{\text{x}\to 1} \frac{2x+3}{3x+2} = \frac{2+3}{3+2} = \frac{5}{5} = 1$

2) $\lim\limits_{\text{x}\to\infty} \frac{2x+3}{3x+2} = \left(\frac{\infty}{\infty}\right) = \lim\limits_{\text{x}\to\infty} \frac{2+\frac{3}{x}}{3+\frac{2}{x}} = \frac{2}{3}$

3) $\lim\limits_{\text{x}\to 2} \frac{2x^2-3x+1}{2x^2+5x-3} = \frac{2\cdot 4-3\cdot 2+1}{2\cdot 4+5\cdot 2-3} = \frac{3}{15} = \frac{1}{5}$

4) $\lim\limits_{\text{x}\to\frac{1}{2}} \frac{2x^2-3x+1}{2x^2+5x-3} = \left(\frac{0}{0}\right) = \lim\limits_{\text{x}\to\frac{1}{2}} \frac{(x-1)(2x-1)}{(x+3)(2x-1)} = \lim\limits_{\text{x}\to\frac{1}{2}} \frac{x-1}{x+3} = \frac{\frac{1}{2}-1}{\frac{1}{2}+3} = -\frac{1}{7}$

5) $\lim\limits_{\text{x}\to\infty} \frac{2x^2-3x+1}{2x^2+5x-3} = \left(\frac{\infty}{\infty}\right) = \lim\limits_{\text{x}\to\infty} \frac{2-\frac{3}{x}+\frac{1}{x^2}}{2+\frac{5}{x}-\frac{3}{x^2}} = \frac{2}{2} = 1$

6) $\lim\limits_{\text{x}\to\infty} \frac{2x+\sin 3x}{3x+\cos 2x} = \left(\frac{\infty}{\infty}\right) = \lim\limits_{\text{x}\to\infty} \frac{2+\frac{\sin 3x}{x}}{3+\frac{\cos 2x}{x}} = \frac{2}{3}$

7) $\lim\limits_{x\to+\infty}\left(\sqrt{4x+3}-\sqrt{4x+1}\right)=(\infty-\infty)=\lim\limits_{x\to+\infty}\frac{(4x+3)-(4x+1)}{\sqrt{4x+3}+\sqrt{4x+1}}=$

$=\lim\limits_{x\to+\infty}\frac{2}{\sqrt{4x+3}+\sqrt{4x+1}}=0$

8) $\lim\limits_{x\to+\infty}\left(\sqrt{4x^2+3x}-\sqrt{4x^2-3x}\right)=(\infty-\infty)=\lim\limits_{x\to+\infty}\frac{(4x^2+3x)-(4x^2-3x)}{\sqrt{4x^2+3x}+\sqrt{4x^2-3x}}=$

$=\lim\limits_{x\to+\infty}\frac{6x}{\sqrt{4x^2+3x}+\sqrt{4x^2-3x}}=\left(\frac{\infty}{\infty}\right)=\lim\limits_{x\to+\infty}\frac{6}{\sqrt{4+\frac{3}{x}}+\sqrt{4-\frac{3}{x}}}=\frac{6}{2+2}=\frac{6}{4}=\frac{3}{2}$

9) $\lim\limits_{x\to-\infty}\left(\sqrt{4x^2+3x}-\sqrt{4x^2-3x}\right)=(\infty-\infty)=\lim\limits_{x\to-\infty}\frac{(4x^2+3x)-(4x^2-3x)}{\sqrt{4x^2+3x}+\sqrt{4x^2-3x}}=$

$=\lim\limits_{x\to-\infty}\frac{6x}{\sqrt{4x^2+3x}+\sqrt{4x^2-3x}}=\left(\frac{\infty}{\infty}\right)=\lim\limits_{x\to-\infty}\frac{-6}{\sqrt{4+\frac{3}{x}}+\sqrt{4-\frac{3}{x}}}=-\frac{3}{2}$

10) $\lim\limits_{x\to+\infty}\frac{2^x+3^x}{2^x-3^x}=\lim\limits_{x\to+\infty}\frac{(\frac{2}{3})^x+1}{(\frac{2}{3})^x-1}=-1$

11) $\lim\limits_{x\to-\infty}\frac{2^x+3^x}{2^x-3^x}=\lim\limits_{x\to-\infty}\frac{1+(\frac{3}{2})^x}{1-(\frac{3}{2})^x}=1$

12) $\lim\limits_{x\to9}\frac{\sqrt{x}-3}{x^2-10x+9}=\left(\frac{0}{0}\right)=\lim\limits_{x\to9}\frac{x-9}{(\sqrt{x}+3)(x-9)(x-1)}=\lim\limits_{x\to9}\frac{1}{(\sqrt{x}+3)(x-1)}=\frac{1}{6\cdot8}=\frac{1}{48}$

13) $\lim\limits_{x\to2}\frac{\sqrt{4x+1}-3}{\sqrt{7x+2}-4}=\left(\frac{0}{0}\right)=\lim\limits_{x\to2}\frac{(4x+1-9)(\sqrt{7x+2}+4)}{(\sqrt{4x+1}+3)(7x+2-16)}=\lim\limits_{x\to2}\frac{4(x-2)(\sqrt{7x+2}+4)}{7(x-2)(\sqrt{4x+1}+3)}=\frac{4\cdot8}{7\cdot6}=\frac{16}{21}$

14) $\lim\limits_{x\to8}\frac{\sqrt[3]{x}-2}{x^2-9x+8}=\left(\frac{0}{0}\right)=\lim\limits_{x\to8}\frac{x-8}{((\sqrt[3]{x})^2+2\sqrt[3]{x}+4)(x-1)(x-8)}=\lim\limits_{x\to8}\frac{1}{((\sqrt[3]{x})^2+2\sqrt[3]{x}+4)(x-1)}=$

$=\frac{1}{12\cdot7}=\frac{1}{84}$

15) $\lim\limits_{x\to2}\left(\frac{1}{x-2}-\frac{12}{x^3-8}\right)=(\infty-\infty)=\lim\limits_{x\to2}\frac{x^2+2x+4-12}{(x-2)(x^2+2x+4)}=\lim\limits_{x\to2}\frac{x^2+2x-8}{(x-2)(x^2+2x+4)}=$

$$= \lim_{x\to 2}\frac{(x-2)(x+4)}{(x-2)(x^2+2x+4)} = \lim_{x\to 2}\frac{(x+4)}{(x^2+2x+4)} =$$

$$= \frac{6}{12} = \frac{1}{2}$$

16) $\lim_{x\to\infty}\left(\arccos\left(\frac{3x+5}{6x+7}\right)\right) = \arccos\left(\frac{1}{2}\right) = \frac{\pi}{3}$

17) $\lim_{x\to\infty}\left(x\sqrt[3]{8x^3+3x+1} - 2x^2\right) = (\infty - \infty) =$

$$= \lim_{x\to\infty}\frac{x(8x^3+3x+1-8x^3)}{\left(\sqrt[3]{8x^3+3x+1}\right)^2 + 2x\sqrt[3]{8x^3+3x+1} + 4x^2} =$$

$$= \lim_{x\to\infty}\frac{3x^2+x}{\left(\sqrt[3]{8x^3+3x+1}\right)^2 + 2x\sqrt[3]{8x^3+3x+1} + 4x^2} =$$

$$= \lim_{x\to\infty}\frac{3+\frac{1}{x}}{(\sqrt[3]{8+\frac{3}{x^2}+\frac{1}{x^3}})^2 + 2\sqrt[3]{8+\frac{3}{x^2}+\frac{1}{x^3}} + 4} = \frac{3}{12} = \frac{1}{4}$$

18) $\lim_{x\to\infty}\left(\frac{2x+5}{2x-3}\right)^x = \lim_{x\to\infty}\left(1+\frac{8}{2x-3}\right)^x = (1^\infty) = \lim_{x\to\infty}(\left(1+\frac{8}{2x-3}\right)^{\frac{2x-3}{8}})^{\frac{8x}{2x-3}} = e^4$

19) $\lim_{x\to 0}(\cos 2x)^{\frac{1}{(\sin x)^2}} = (1^\infty) = \lim_{x\to 0}(1-2\,(\sin x)^2)^{\frac{1}{(\sin x)^2}} =$

$$= \lim_{x\to 0}((1-2\,(\sin x)^2)^{\frac{1}{-2\,(\sin x)^2}})^{-2} = e^{-2}$$

20) $\lim_{x\to 1}\left(\frac{5x+2}{6x+1}\right)^{\frac{1}{x-1}} = (1^\infty) = \lim_{x\to 1}\left(1+\left(\frac{5x+2}{6x+1}-1\right)\right)^{\frac{1}{x-1}} =$

$$= \lim_{x\to 1}(\left(1-\frac{x-1}{6x+1}\right)^{-\frac{6x+1}{x-1}})^{-\frac{1}{6x+1}} = e^{-\frac{1}{7}}$$